

AWS re:Invent

NOV. 28 – DEC. 2, 2022 | LAS VEGAS, NV

DOP203-R

Amazon's DevOps culture

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Introductions



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What we will learn today

- How Amazon builds organizations, processes, and systems that enable rapid innovation and delivery of high-performing digital solutions for its customers
- Lessons learned at Amazon for developing and delivering software
- Amazon's best practices for building, testing, deploying, and maintaining modern applications
- Q&A throughout

Lessons learned at Amazon for developing and delivering software



Decompose for agility
(**microservices, two-pizza teams**)



Automate everything



Standardized tools



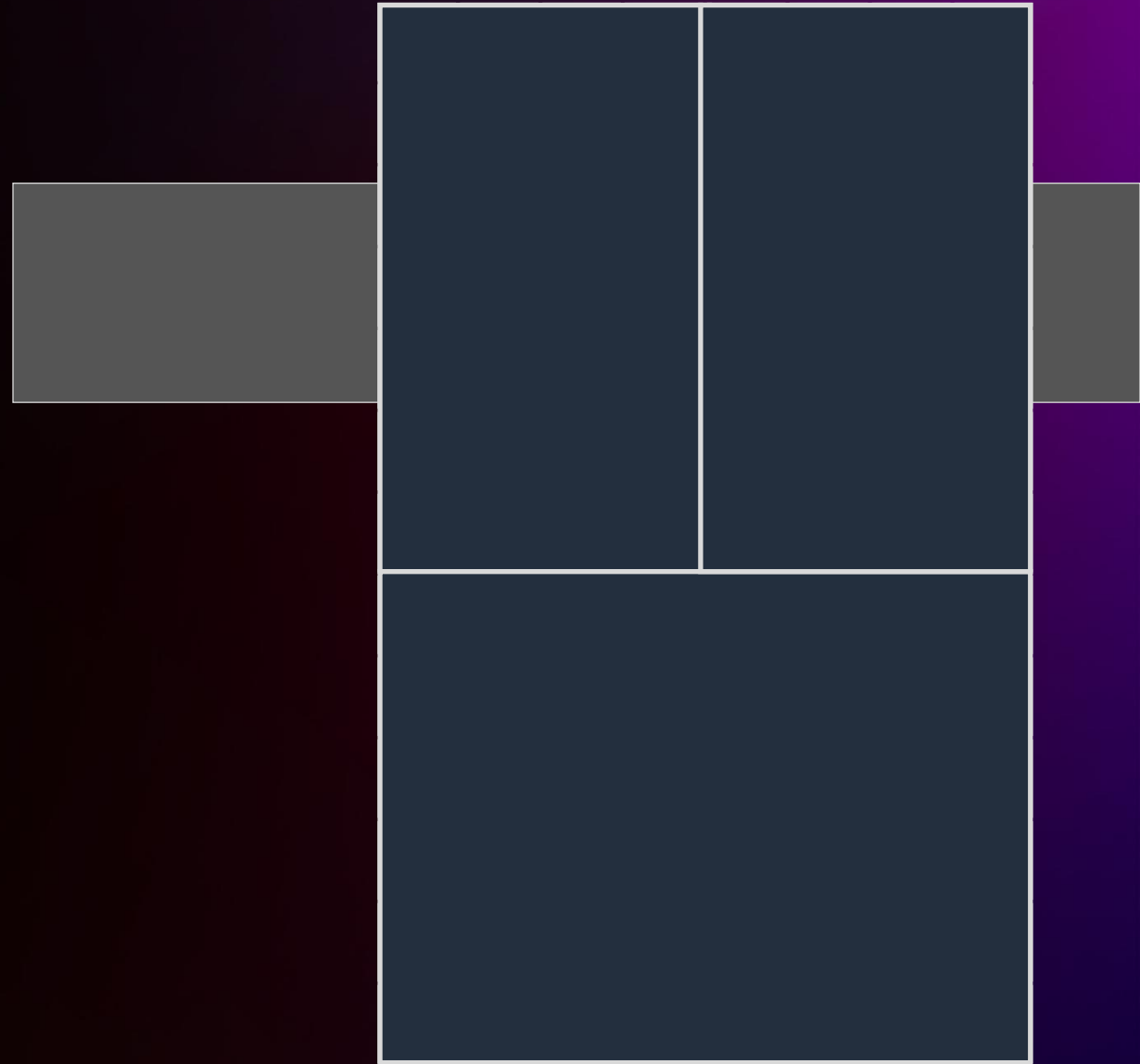
Belts and suspenders
(**governance, templates**)



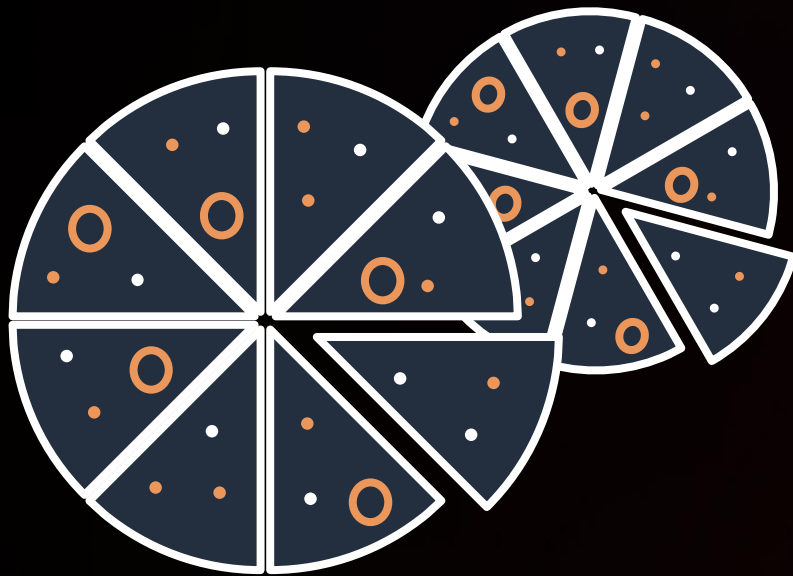
Infrastructure as code

Decompose for agility

- Principles
- Make units as small as possible (primitives)
- Decouple based on scaling factors, not functions
- Each service operates independently
"Communication is terrible!" – Jeff Bezos
- APIs (contracts) between services



Getting (re)organized



“Two-pizza” teams

- Own a service
- Minimize social constraints (Conway’s law)
- Autonomy to make decisions

Teams own everything

- Planning
- Security
- Performance
- Scalability
- Deployment
- Operation
- Bugs
- Documentation
- Testing



How can one team do all of this?

Modern applications

Today we have modern applications that



- Use independently scalable microservices (serverless, containers)
- Connect through APIs
- Deliver updates continuously
- Adapt quickly to change
- Scale globally
- Are fault tolerant
- Carefully manage state and persistence
- Have built-in security

Measuring success

Business metrics	Operational metrics	Input goals	Enablement
Growth	Errors	Features	Principal reviews
Usage	Throttling	Use cases	Security training
Feedback	Failed deployments	Performance	Ops training
	Performance		

Q&A



**I want my organization to
build modern applications –
where do I get started?**

**I want to move from a monolith
to a microservices architecture –
where do I begin?**

How do I maintain security and compliance in my production CI/CD pipelines while giving my developers the ability to innovate and be agile in their development environments?

How do I restructure my teams for building modern applications?

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Thank you!

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